

RESEARCH ARTICLE

Refining classification of complex agroforestry mosaic landscape from drone-based imagery: Implications for landscape management and conservation

Leslie Annette Paige¹  | Katherine Hedger^{2,3} | Esther Adinda² | Endah Sulistyawati⁴ | Alexander S. Bradley⁵ | Bill Winston⁵ | Karen E. DeMatteo⁶  | K. A. I. Nekaris^{2,3}  | Emily E. Wroblewski¹

¹Department of Anthropology, Washington University in St. Louis, Saint Louis, Missouri, USA; ²Little Fireface Project, Garut, Java, Indonesia; ³School of Life Sciences, Centre for Ecology, Evolution and Environment, Anglia Ruskin University, Cambridge, UK; ⁴Institut Teknologi Bandung, Bandung, Java, Indonesia; ⁵Department of Earth, Environmental and Planetary Sciences, Washington University in St. Louis, Saint Louis, Missouri, USA and ⁶Department of Biology, Washington University in St. Louis, Saint Louis, Missouri, USA

Correspondence

Leslie Annette Paige
Email: leslie.p@wustl.edu

K. A. I. Nekaris
Email: anekaris@littlefireface.org

Funding information

Augsburg Zoo; International Primate Protection League; Omaha's Henry Doorly Zoo; Washington University in St. Louis; People's Trust for Endangered Species; Cleveland Metroparks Zoo

Handling Editor: Javier Lopatin

Abstract

1. Accurately and efficiently mapping landscape compositions is a conservation priority as natural habitats that are reservoirs of biodiversity are increasingly lost or modified. Globally, agroforestry is a rising sustainable land-use type that benefits landowners and biodiversity, but these land-use land-cover (LULC) mosaics can be challenging to map due to the size and distribution of different LULC patches, requiring improved methodologies to map these areas and refine landscape-based conservation management strategies.
2. We classified an agroforestry landscape in West Java, Indonesia, which includes the habitat of a Critically Endangered primate, the Javan slow loris (*Nycticebus javanicus*). We used object-based image analysis of high-resolution drone (DJI Phantom 4 Pro, 2.90 cm/px mean resolution) and satellite (PlanetLabs, 3 m resolution) imagery, comparing how the resulting classifications differ and impact the classification of GPS waypoints collected during focal animal follows.
3. Estimations of LULC types significantly differed between the classified drone and satellite imagery, affecting all habitat types. Compared to classifications from drone imagery, satellite imagery-based classifications overestimated forest cover by an average of 18.81% and chayote (a vining crop) by 16.02%, while underestimating non-chayote agriculture by 18.72%. Specific crop types were also classified from drone imagery that were impossible to classify using satellite imagery.
4. Image classification differences are reflected in significantly different classifications of loris GPS waypoints, with an overestimation of the amount of waypoints in forest and chayote, and underestimation of waypoints in non-chayote agriculture.

This is an open access article under the terms of the [Creative Commons Attribution](https://creativecommons.org/licenses/by/4.0/) License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2026 The Author(s). *Journal of Applied Ecology* published by John Wiley & Sons Ltd on behalf of British Ecological Society.

5. *Synthesis and applications.* Together, these results show that significant differences in landscape classification between high-resolution satellite and drone imagery can substantially change our understanding of slow loris habitat availability and use. Thus, it can be beneficial to use the highest resolution imagery to characterize species' habitats, whenever possible, particularly within mosaic landscapes. Additionally, image and map resolution should be explicitly reported. Less accurate area estimations of LULC types, particularly for forest and agriculture, directly impact conservation planning because species persistence may rely on these habitat patches and restoration of connectivity between them. Changes to forest and agriculture LULC can reflect the socio-economic processes driving landscape changes.

KEYWORDS

drone imagery, satellite imagery, remote-sensing, object-based image analysis, Javan slow loris, habitat mapping and availability, habitat use

1 | INTRODUCTION

Globally, human-induced land cover change is occurring at rapid rates, with immense consequences for a myriad of ecological, environmental and economic processes (Foley et al., 2005), particularly with loss of biodiversity contributing to local and global climate change and ecosystem disruption (Winkler et al., 2021). Land cover is changing largely via large-scale deforestation, urbanization and agriculture expansion that reduces natural habitats (Lambin & Meyfroidt, 2011; Radwan et al., 2021). The outcomes are landscapes that are fragmented and composed of complex mixtures of land-use types (referred to as heterogeneous or mosaic landscapes), which pose a large threat to the persistence of global biodiversity (Durán et al., 2020; Newbold et al., 2015). With growing human populations and demand for resources, rates of land conversion and associated biodiversity loss are projected to increase in coming years (Jacobson et al., 2019; Semenchuk et al., 2022).

A growing global practice to mitigate the impacts of habitat loss is the adoption of sustainable landscape management plans that benefit both human and wildlife populations. Agroforestry is a sustainable mosaic landscape, where trees and shrubs are grown alongside crops or pastures, providing economic benefits for landowners and useable habitats for native wildlife (Nair et al., 2021). Monitoring and mapping patterns of agroforestry and other land uses is a high priority for stakeholders since designing effective management strategies requires an accurate understanding of the landscape (Foley et al., 2005; Turner 2nd et al., 2007). Satellite imagery is often used to classify landscapes, and the variation in resolution of available satellite imagery may mean there is sufficient resolution to assess landscape heterogeneity and species' needs in some circumstances. However, many mosaic landscapes are challenging to accurately map as they vary on spatial scales smaller than can be captured with satellite imagery, and their different components have high spectral similarity (Gibril et al., 2020;

Hurskainen et al., 2019; Mercier et al., 2019). Consequently, smaller but ecologically significant patches or corridors of vegetation existing outside of larger forest tracts, like in agroforestry landscapes, are often excluded from vegetation estimates (Lechner et al., 2009; Räsänen & Virtanen, 2019; Reiner et al., 2023). It is necessary to quantify these features to track and improve the predictions of global land-use trends and facilitate wildlife persistence in modified landscapes (Lindenmayer, 2019; Wintle et al., 2019; Yan et al., 2021).

Imagery from unmanned aerial vehicles (UAVs or drones) has great potential to be incorporated into landscape monitoring and wildlife habitat mapping as UAVs can capture imagery at a higher resolution than satellites, down to cm/px (Koh & Wich, 2012). However, UAV imagery has been minimally used to map animal habitats (Han et al., 2017; Stark et al., 2018; Szantoi et al., 2017). Improving the accuracy and specificity of land cover classification can change our estimation of the total area and distribution of land cover types, which, in turn, can change our understanding of how endangered species use habitats, as well as how best to protect the species and the habitats they depend on. Therefore, this study investigates whether drone-based imagery improves classifications of a highly human-modified, complex, agroforestry mosaic landscape in Java (Indonesia).

1. We compared land-use land-cover (LULC) classifications from high-resolution drone (mean resolution of 2.87 cm/px) and high-resolution satellite imagery (3 m). Java is an island experiencing rapid rates of land conversion while having one of the highest human population densities in the world (Jones, 2013; Pravitasari et al., 2024). Agroforestry is the dominant land-use type across the island, followed by monoculture crops, making refining the ability to classify these landscapes important for the development of conservation strategies for Java's vulnerable wildlife populations (Purwaningsih et al., 2020; Udawatta et al., 2021).

2. We then tested whether differences in classification from the two image sources change classifications of GPS waypoints collected during focal animal follows of the Critically Endangered Javan slow loris (*Nycticebus javanicus*) because GPS waypoints, and associated LULC characteristics, are often incorporated in analyses of animal–habitat relationships. The Javan slow loris is a strictly arboreal primate that is sensitive to fine scale landscape variation because individuals have relatively small home ranges and rely on vegetation connectivity to navigate the landscape (Nekaris et al., 2017). Loss of forest habitat has contributed to the Javan slow loris becoming one of the 25 most endangered primates in the world (Mittermeier et al., 2022). However, like many species on Java, lorises persist by using human-modified landscapes that are adjacent to forest habitats, which makes classifying these complex, dynamic landscapes, as well as understanding how species use or avoid them, priorities to promote and maximize human–biodiversity coexistence (Nekaris et al., 2017).

2 | METHODS

2.1 | Study site and loris population

Research was conducted at the Little Fireface Project (LFP) field research site in Cipaganti, Cisurupan, Garut Regency, West Java, Indonesia (Figure 1a,b). Drone surveys were deemed exempt by Washington University in St. Louis' Institutional Animal Care and Use Committee (IACUC) and conducted under research

permit number 365/SIP/IV/FR/6/2023 from Indonesia's National Research and Innovation Agency. Javan slow loris behavioural research was conducted according to the guidelines of the Declaration of Helsinki and followed the American Society of Primatologists Principles for the Ethical Treatment of Non-human Primates. It was approved by the Animal Ethics Subcommittee of Oxford Brookes University permit number 12OXASA2724. The landscape is a mosaic of human development, agriculture, and trees that exist either in rows to demarcate fields or in scattered patches (Nekaris et al., 2017). The main source of income for Cipaganti residents is agriculture, with available land used for that purpose. Dominant crops include coffee (*Coffea arabica*), tea (*Camellia sinensis*), tomato (*Solanum lycopersicum*) and chayote (*Sechium edule*) (Nekaris et al., 2017). Chayote, a vining crop grown on elevated bamboo lattice frames, provides substrate that the strictly arboreal Javan slow loris can traverse (Nekaris et al., 2017) (Figure 2).

Javan slow lorises live in single male–single female family groups, which may include 1–4 offspring (Nekaris et al., 2020). This nocturnal species is highly territorial, with venom used for defence, including during intraspecific competition; consequently, neighbouring family groups have very little home range overlap (Nekaris et al., 2020) (Figure 1c). One year (August 2022–July 2023) of ranging data, collected by LFP, included nine loris family groups, totalling 16 lorises. Focal individuals were followed nightly (1700–0500h), six nights a week and GPS waypoints were collected using a handheld Garmin unit at 5-min intervals (Biot et al., 2020).

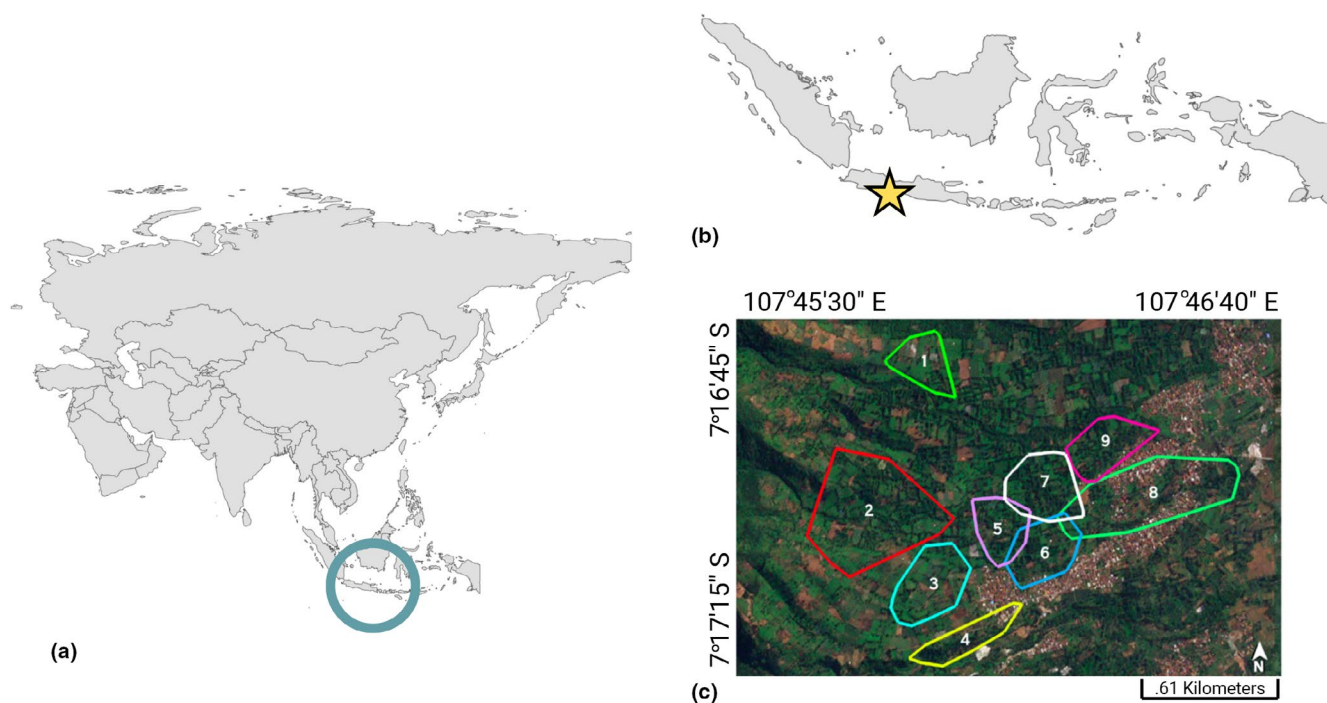


FIGURE 1 (a) Location of the island of Java, Indonesia. (b) Location of Cipaganti in West Java. (c) One-year, 100% minimum convex polygon home ranges of nine loris family groups at the study site in Cipaganti overlaid on 3m satellite imagery from 29 July 2023. Panels (a) and (b) © RNaturalEarth; Image (c) © 2023 Planet Labs PBC.



FIGURE 2 Chayote (*Sechium edule*) is a vining crop grown on elevated bamboo lattice frames (left, © Leslie Annette Paige) that provides the only elevated substrate outside of forest that the arboreal Javan slow loris can traverse (middle). This contrasts their natural forest habitat (right) (middle, right © Little Fireface Project).

2.2 | Drone and satellite imagery acquisition

2.2.1 | Drone surveys

GPS data collected during focal animal follows over the preceding year defined the 100% minimum convex polygon home ranges surveyed with a DJI Phantom 4 Pro drone in July 2023 (Figure 1c). We conducted eight surveys, totalling 117 ha, from 18 to 29 July, over the nine loris family group home ranges (ranges 5 and 6 surveyed together). Prior to each survey, an Emlid Reach RS2 GNSS unit in the WGS 84/UTM Zone 48S reference system recorded six ground control points (GCPs) per survey area, ensuring accurately georeferenced drone imagery (Martínez-Carricondo et al., 2018) (see Supporting Information for more detail).

2.2.2 | Satellite imagery

We acquired 3m resolution satellite imagery of Cipaganti from Planet Labs (Planet Team, 2017). Imagery was captured with the PlanetScope constellation of DOVE satellites. Although the satellites take daily images, we used images from 29 July 2023, the closest day to the surveys without cloud cover. Imagery was downloaded with analytic radiance (TOAR) to adjust for atmospheric radiance, resulting in four bands. Only RGB bands were analysed to be comparable with the drone imagery.

2.3 | Data processing

GCPs were post-processed using the Canadian Spatial Reference System Precise Point Positioning (NRCan-PPP), which improves spatial positioning (Natural Resources Canada, 2017). We processed drone imagery using Agisoft Metashape, in which individual photographs were aligned to create a single, unified orthomosaic map of the survey area (Gibril et al., 2020; Kamarudin & Wei, 2022). We then georeferenced the map by manually entering and identifying GCP coordinates.

2.4 | Data analyses

2.4.1 | Machine learning landscape classification

We used object-based image analysis (OBIA) in Catalyst Professional to classify landscape types in the drone orthomosaic maps and satellite imagery. Briefly, OBIA classifies imagery based on pixel groups, or discreet 'objects', grouped together based upon spectral, shape and texture variables (Blaschke, 2010; Hossain & Chen, 2019) (see Supporting Information for more detail). Satellite imagery was classified as forest, chayote, non-chayote agriculture and developed land (e.g. housing and other structures) (Figure 3). Non-chayote agriculture was further distinguished in the drone imagery as row crops (e.g. chilies, tomatoes, cabbages) or dot crops (e.g. coffee, tea).

Image classification accuracy was assessed using a confusion matrix, which compares locations on the ground of known use ('ground truth') to the image classification (Manohar et al., 2021). Here, drone imagery classification was used as the ground truth to which satellite classification was compared using 100 randomly stratified accuracy assessment points in each of the nine loris home ranges (Li et al., 2024; Quattrini et al., 2025), following the recommended 50–100 assessment points per LULC class (Congalton, 2001; Foody, 2002).

2.4.2 | Loris home range and GPS waypoint classification comparison

We defined 100% minimum convex home range polygons for each of the nine loris family groups using the Minimum Bounding Geometry tool in ArcGIS and all GPS waypoints from August 2022 to September 2023, given that this species is highly territorial with stable home ranges. The resulting polygons were overlaid on the classified habitat type maps in ArcGIS, and full maps clipped to the home range areas for analyses. For each home range, the area of each LULC type was calculated from the classified satellite and drone imagery. To compare classification proportions of GPS waypoints collected during

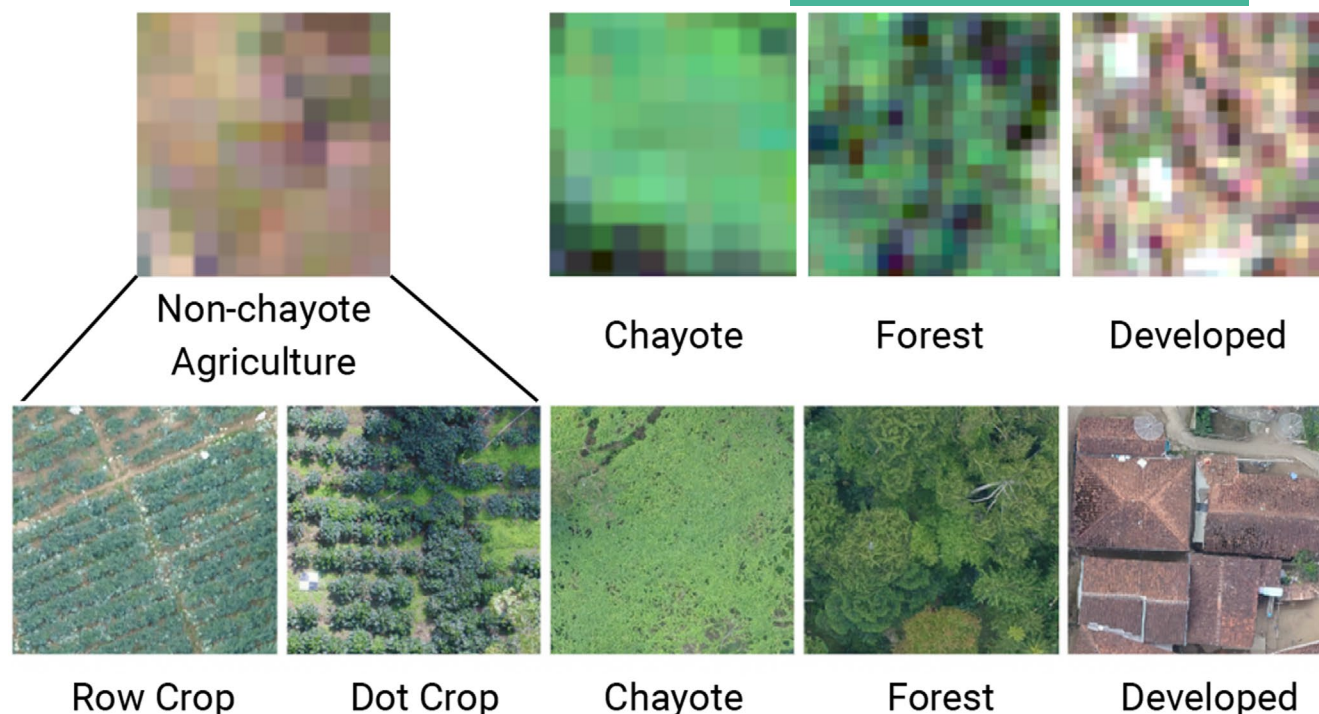


FIGURE 3 Representative images of discernible land-use land-cover (LULC) classifications from 3m satellite imagery (top, © Planet Labs PBC) and DJI Phantom 4 Pro drone surveys (bottom). Non-chayote agricultural land identified via satellite imagery was classified into row and dot crops using drone imagery.

loris focal follows, the waypoints (total=32,142, mean=3571 points per home range [range: 1901–7202]) were overlaid on both the classified drone and satellite maps, and the LULC type for each point was extracted. We excluded points that fell on borders between LULC types.

2.4.3 | Statistical analyses

We used Chi-squared and Fisher's exact tests to test for differences in home range LULC classification and loris waypoint classifications based on imagery source. Drone-based classifications of row and dot crop were combined and analysed as 'non-chayote agriculture' to allow comparison with satellite imagery (Hsu et al., 2020). Analyses used the total, aggregated data from all loris home ranges and were conducted in R 4.4.2.

3 | RESULTS

3.1 | Orthomosaic maps

The drone collected 2970 photos over the eight surveys covering nine loris home ranges (mean survey area=13 ha; range: 6–30 ha), of which 2915 images were successfully aligned (98.14%) to generate eight orthomosaic maps (Figure 4). Mean map image resolution among the eight survey areas was 2.87 cm/px (range: 2.28–4.31 cm/px).

3.2 | Satellite and drone imagery LULC classification

LULC classification significantly differed between satellite and drone imagery (χ^2 [3, $N=18$]=2559.80, $p<0.005$) and affected all LULC types (Fisher's exact tests, all comparisons $p<0.005$) (Table 1; Figure 4; Figure S1; Table S1). Satellite classification overestimated forest in six of nine loris ranges by a mean of 18.81% (range: 4.88%–37.35%) (Figures 5 and 6). Satellite classification estimated forest comprised a mean of 47.07% of loris home ranges, compared to 36.06% from drone classification. Similarly, chayote was overestimated in six of nine ranges by a mean of 16.02% (range: 2.98%–42.98%), with chayote accounting for a mean of 35.15% of loris range area from satellite classification but 25.54% from drone classification. Satellite classification underestimated non-chayote agriculture in eight of nine ranges by a mean of 16.76% (range: 4.87%–25.38%), with non-chayote agriculture comprising a mean of 12.41% of loris ranges for satellite classification but 32.72% from drone classification. In all ranges, the total non-chayote agricultural area classified from satellite imagery was less than that identified via drone classification, suggesting that drone-based classification more accurately identifies total agricultural land (in addition to chayote), regardless of the ability to differentiate between row and dot crops. Developed land was the least over- or underestimated from satellite imagery (overestimation range: 0.14%–5.45%, underestimation range: 0.21%–3.49%), but these estimated differences are notable given developed land consistently comprised a low proportion of the home range (0%–5.47%), except in Group 8 (satellite: 29.66%; drone: 31.68%).

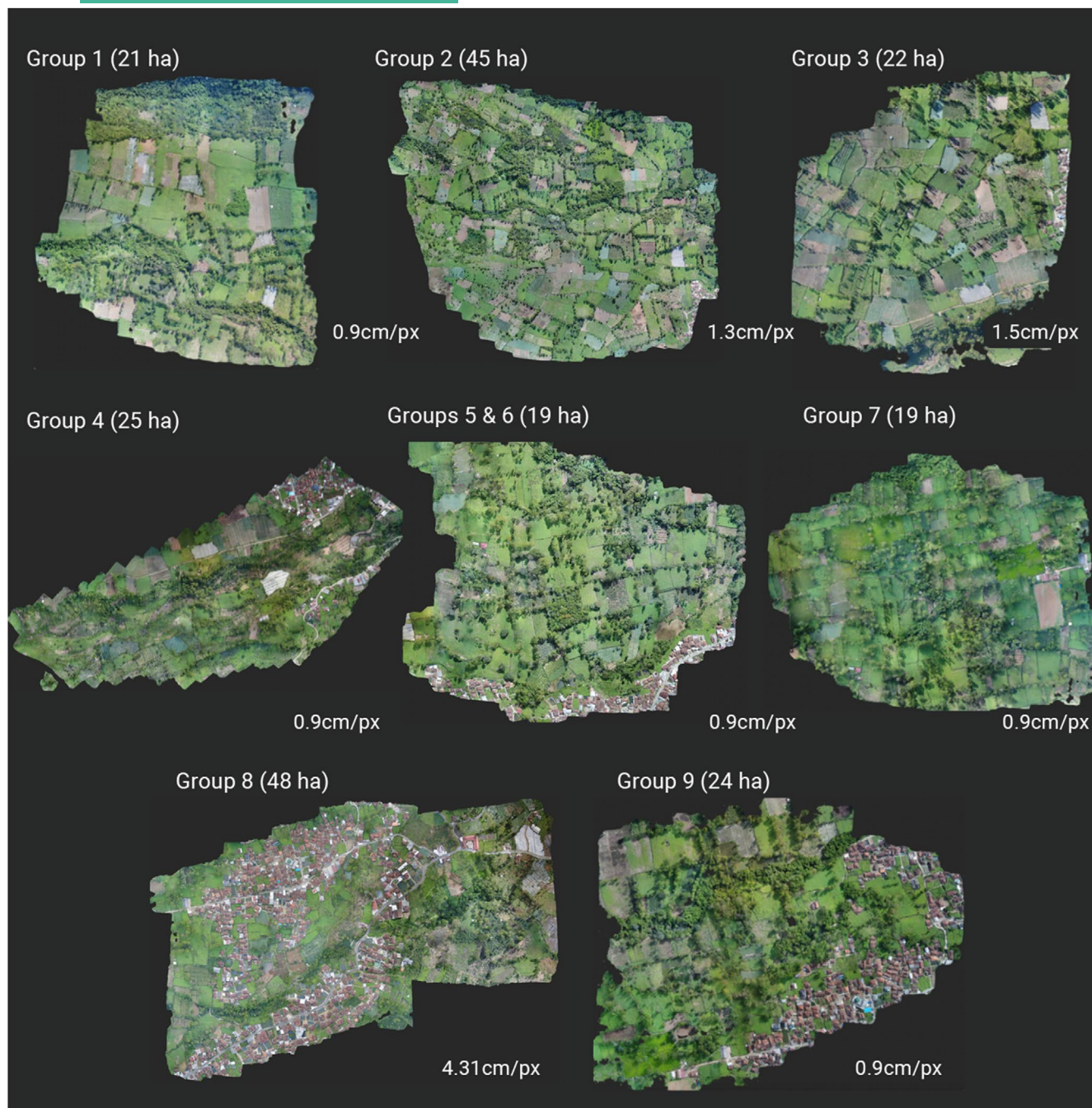


FIGURE 4 Orthomosaic maps generated from the drone surveys of the loris home ranges. Group numbers correspond to those in Figure 1, with Groups 5 and 6 surveyed together. For each group and survey, the image resolution (cm/px) and total area (ha) are noted.

Confusion matrices indicated a low kappa coefficient (overall accuracy) of the satellite compared to drone classifications (mean = $27.77\% \pm \text{SD } 11.07\%$ agreement between the two classifications) (Figure 7; Tables S2–S10). Forest was classified most accurately (mean = $66.11\% \pm \text{SD } 22.12\%$), followed closely by chayote (mean = $58.55\% \pm \text{SD } 21.25\%$), but both still had a low overall accuracy ($<70\%$). Satellite imagery classified non-chayote agricultural and developed land with $<30\%$ accuracy (agriculture: mean = $26.33\% \pm \text{SD } 15.00\%$; developed: mean = $26.12\% \pm \text{SD } 29.52\%$). Agriculture was

most often misclassified as developed area, forest as non-chayote agriculture, developed land as chayote and chayote as non-chayote agriculture.

3.3 | Classification of loris GPS waypoints

The proportion of GPS waypoints collected during loris focal follows that associated with different LULC types differed between

TABLE 1 Chi-squared and Fisher's pairwise statistical comparisons for total LULC area (ha) determined from drone and satellite imagery classification. Additional Chi-squared statistical output is provided in [Table S1](#).

Chi-squared					
<i>p</i> -value		χ^2	Degrees of freedom		
<i>p</i> < 0.005		2559.80	3		
Fisher's pairwise					
LULC	Satellite area estimate (%)	Drone area estimate (%)	<i>p</i> -value	95% Confidence interval	Odds ratio
Forest	41.02	32.65	<i>p</i> < 0.005	0.673–0.722	0.697
Chayote	37.66	25.20	<i>p</i> < 0.005	0.538–0.578	0.558
Non-chayote agriculture	13.73	32.05	<i>p</i> < 0.005	2.822–3.114	2.964
Developed	7.59	10.09	<i>p</i> < 0.005	1.280–1.457	1.366

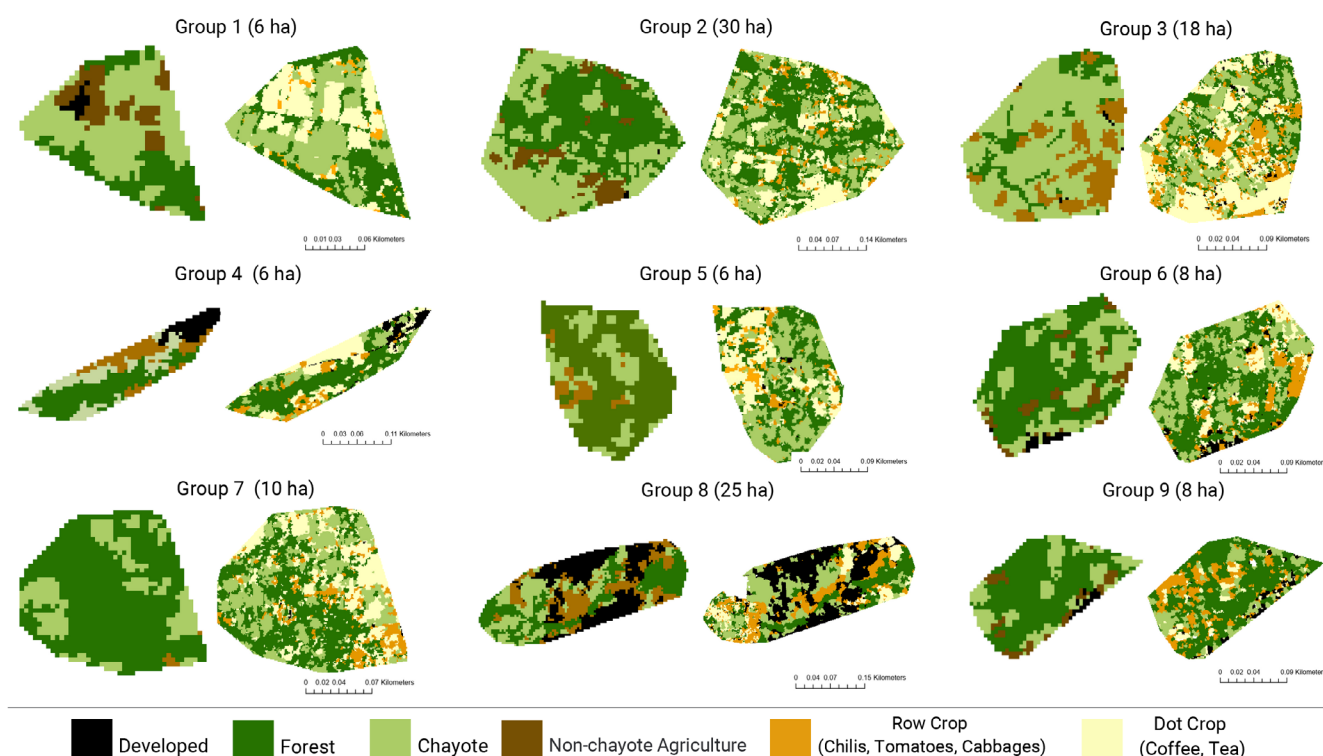


FIGURE 5 Classified maps for each slow loris home range derived from Planet Labs satellite imagery (left image) and DJI Phantom 4 Pro drone imagery (right image), with area (ha) of each loris home range. Row and dot croplands were distinguishable via drone imagery, whereas only general, non-chayote agriculture was distinguishable via satellite imagery. Images not to scale.

the classification derived from satellite and drone imagery (χ^2 [3, $N=18$]=3135.70, $p < 0.005$) and affected all LULC types (Fisher's exact tests, all comparisons $p < 0.005$) ([Table 2](#); [Figure 5](#); [Figure S2](#)). Consistent with the LULC classification differences between satellite and drone imagery, loris habitat use of forest and chayote was overestimated while use of non-chayote agriculture was underestimated.

4 | DISCUSSION

LULC area estimations differed between satellite versus drone imagery classifications in the complex agroforestry mosaic landscape in Cipaganti, Java, Indonesia. Although we expected improved classification from drone imagery because of its higher resolution compared to the satellite imagery, such differences can change our

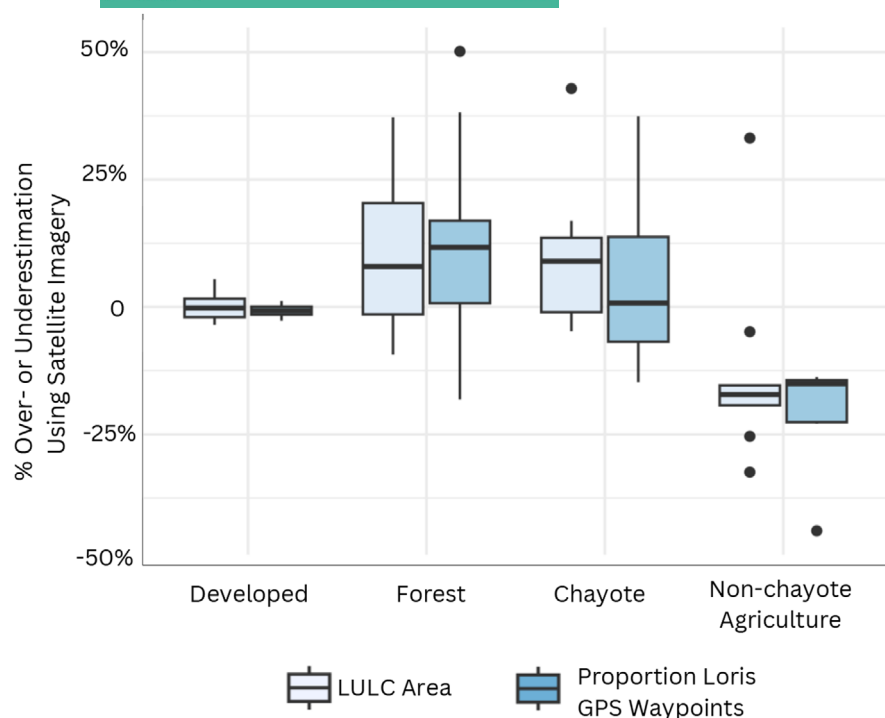


FIGURE 6 Per cent over- or underestimation of the area of each LULC type (light blue) and proportion of loris focal follow GPS waypoints associated with each LULC type (dark blue) when comparing satellite to drone LULC classification among nine loris home ranges. Non-chayote agriculture was overestimated in one range but its use by lorises was not overestimated in any range.

understanding of land use by both wildlife and people as well as resulting management and conservation decisions. Accordingly, these differences in LULC classification resulted in different LULC proportions of the Javan slow loris GPS waypoints, which could impact our understanding of their habitat use and behaviour.

4.1 | Mosaic landscapes: Importance of resolution and causes of misclassification

Globally, forests are one of the most important natural resources, making accurate mapping of forest cover a priority for mitigating climate change and supporting REDD+ (Bonan, 2008; Canadell & Raupach, 2008). Assessments via satellite imagery often fail to fully capture trees existing outside of large forest tracts, which could be due to tree crowns being smaller than what is detectable via the image resolution and/or spectral similarity with other LULC types (Brandt et al., 2020; Liu et al., 2023; Schnell et al., 2015; Thomas et al., 2021). These issues likely contributed to the common overestimation of forest availability within our study area when using high-resolution satellite imagery.

Emerging advanced remote sensing tools and classification methods have been shown to improve forest availability estimates (Illarionova et al., 2024; Lechner et al., 2020; Mugabowindekwe et al., 2023; Reiner et al., 2023). For example, forest estimates can be improved when vegetation indices, such as NDVI, calculated from multi- and hyperspectral imagery bands (e.g. UV, near infrared) are combined with RGB imagery (Pu & Landry, 2012). This study relied on high-resolution, standalone RGB imagery for landscape classification, with a significant benefit of it being more cost-effective

and straightforward to process than other imagery (Onishi & Ise, 2021). Drones also show great potential to monitor vegetation, with the ability to identify individual trees and other plant species (Bertacchi et al., 2019; Cruzan et al., 2016; Weinstein et al., 2021). Consequently, forest monitoring could benefit from incorporating RGB drone surveys to better estimate trees existing in and out of forest tracts (Liu et al., 2023; Zhang et al., 2016).

A 3 m satellite imagery underestimated non-chayote agricultural land, overestimated chayote and could not distinguish additional crop types. This may be due to spectral similarity among crop types and the inability to detect different (earlier) growing stages (Wu et al., 2023). However, high-resolution drone imagery and OBIA enabled finer separation of different agricultural crop types: chayote, dot (coffee and tea) and row crops (chillies, tomatoes and cabbages). Although this higher resolution was critical for distinguishing objects for LULC classification, differences in sensors and processing factors such as satellite imagery atmospheric corrections likely also contributed to classification differences for agriculture or other land uses between drone and satellite imagery. Distinguishing specific crop types versus generically identifying agricultural land is important because crops can have different meanings and values (economic and social), and their different structural characteristics also provide different habitats (Masolele et al., 2024). Additionally, agroforestry is widely considered a more sustainable form of agriculture than monoculture crops because it provides more habitat availability for a variety of species (Santos et al., 2022). While crop monitoring will continue to be a complex topic that has challenges of scalability, incorporating drone surveys can improve local agricultural monitoring (Murugan et al., 2017; Xiang & Tian, 2011).

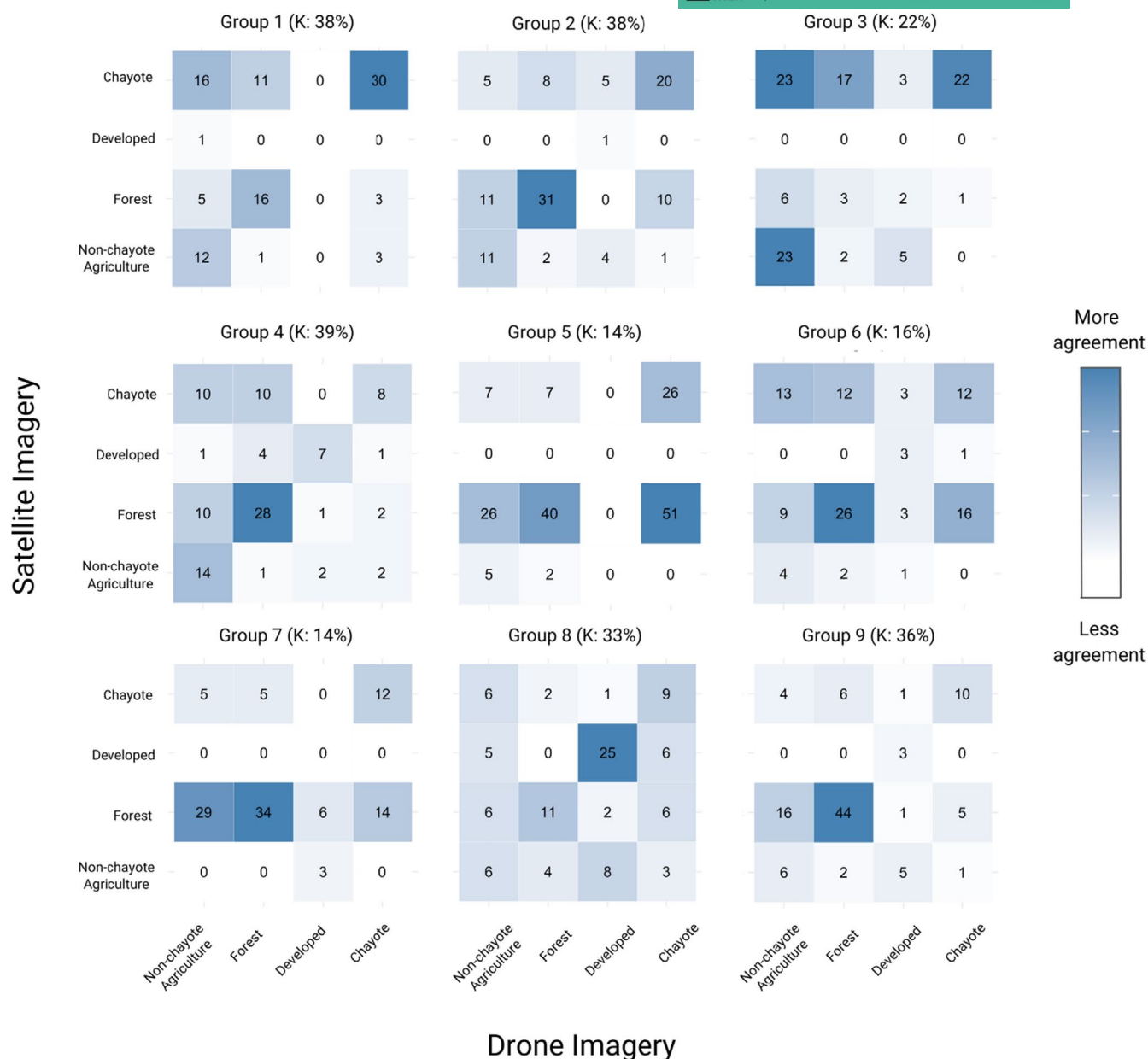


FIGURE 7 Confusion matrices assessing accuracy of satellite imagery LULC classifications within each Javan slow loris home range. Columns represent classifications from drone imagery (used as ground truth), and rows represent satellite imagery classifications. Lighter colours indicate higher classification accuracy (agreement) between the imagery sources. See [Tables S2–S10](#) for additional confusion matrix outputs.

4.2 | Implications for loris conservation

Historically, the Critically Endangered Javan slow loris lived in Indonesia's widespread lowland forests, but <9% of this habitat remains, and it continues to shrink due to extensive human-driven habitat modification (Global Forest Watch, 2023). Remaining loris habitat is mainly mosaic, where small vegetation patches facilitate the persistence of this arboreal species. Distinguishing chayote from other agriculture types is important for slow lorises because the elevated bamboo frames used to grow chayote provide connectivity for lorises to travel within their home ranges. Satellite imagery overestimated the area of forest and chayote, and that

could cause overestimation of their use by the loris, as suggested by differences in waypoint assignments to each LULC type. This overestimation could bias any subsequent behaviour and habitat use analyses. It is vital to refine the Javan slow loris' use of non-chayote agriculture areas since they can exhibit terrestrial behaviour in the absence of tree and/or chayote connectivity, but this puts them at high risk of predation (Karimloo et al., 2023; Nekaris et al., 2024). Refined habitat maps can also prioritize areas for habitat connectivity and installation of other mitigations, such as canopy bridges (Biot et al., 2020).

Many conservation initiatives focus on mitigating the effects of habitat loss and fragmentation through actions that restore

TABLE 2 Chi-squared and Fisher's pairwise statistical comparisons of the proportion of loris focal follow GPS waypoints associated with different LULC types derived from drone versus satellite imagery classification. Additional Chi-squared statistical output is provided in Table S11.

Chi-squared					
<i>p</i> -value		χ^2	Degrees of freedom		
<i>p</i> < 0.005		3135.70	3		
Fisher's pairwise					
LULC	Satellite loris waypoint classifications (%)	Drone loris waypoint classifications (%)	<i>p</i> -value	95% Confidence interval	Odds ratio
Forest	60.59	46.87	<i>p</i> < 0.005	0.555–0.592	0.574
Chayote	25.54	20.71	<i>p</i> < 0.005	0.733–0.790	0.762
Non-chayote agriculture	11.78	29.44	<i>p</i> < 0.005	1.141–1.219	1.180
Developed	2.09	2.97	<i>p</i> < 0.005	1.293–1.588	1.433

habitats, increase areas under protection and promote connectivity (Mokany et al., 2020). In concert with animal behaviour data, landscape data from remote sensing and ground surveys underlie analyses used to determine priority areas for landscape management plans (Schmitt et al., 2009). Therefore, over- or underestimation of LULC, as observed in this study, has important ramifications for how we understand and protect endangered species' habitat. Small vegetation patches are important for protecting global biodiversity as, in many landscapes, patches may be all that remain of the original habitat and serve as footholds for endemic wildlife (Lindenmayer, 2019; Wintle et al., 2019; Yan et al., 2021). Threats faced by many species also occur on localized scales such as small-scale resource extraction, illegal hunting and deforestation (Estrada et al., 2017). Although there is increasing support for using innovative methods of non-invasive habitat monitoring to conserve biodiversity (Koh & Wich, 2012; Piel et al., 2021), drones have been minimally used to classify animal habitats (Han et al., 2017; Stark et al., 2018; Szantoi et al., 2017). The increased accuracy provided by drone-based imagery has great potential to improve habitat assessments, particularly given that many endangered species exist in small pockets within increasingly fragmented and mosaic landscapes (Schaffer-Smith et al., 2016).

5 | CONCLUSION

The persistence of animal and human populations is interrelated, and the path towards conserving biodiversity requires attention to the needs and challenges of both. Because a large driver of global deforestation is land conversion for commodities and agriculture (Curtis et al., 2018), the design of land management strategies that jointly provide resources for human populations and habitats to sustain biodiversity is needed (Weiss et al., 2020). Better quantification of LULC trends can improve the design of sustainability measures, in

partnership with local populations and better support the needs of both people and wildlife.

Drones offer a powerful mapping option and may be particularly effective for monitoring mosaic landscapes where variation occurs on smaller spatial scales than can be captured with satellite imagery. Differences in satellite imagery resolution, including higher available resolutions than featured in this study, may additionally result in classification differences. As satellite imagery can be purchased or freely available, cover large areas with frequent pass rates and is available at a variety of resolutions, drones may not be a replacement but rather a complementary tool that can be used to validate satellite imagery and train models to extrapolate findings to larger areas (Ventura et al., 2022). Drone surveys also have considerations such as reducing disturbance to wildlife, appropriate ground survey design, survey planning to minimize sources of image distortion and error that can be introduced via flight parameters (e.g. height), camera specifications and settings (e.g. image overlap), aerial imagery (e.g. view-angle distortion, parallax, relief displacement) and orthomosaic generation (e.g. stitching artefacts) (Seifert et al., 2019; Young et al., 2022). Selecting both imagery source (drone or satellite) and resolution will vary based upon the specific characteristics of the study landscape and wildlife in question. As classified landscape maps are taken at face value as accurate assessments and reused for a variety of disciplines, spatial error or classification discrepancies present within the maps can have large 'downstream effects' for any analyses that rely on those data (Estes et al., 2018). These discrepancies can have real-world implications for landscape management decisions. Therefore, it is essential to report the spatial resolution of imagery used, as well as any processing techniques that may introduce additional error (Lechner et al., 2012).

AUTHOR CONTRIBUTIONS

Leslie Annette Paige, K. A. I. Nekaris, and Emily E. Wroblewski conceived the ideas. Leslie Annette Paige, Alexander S. Bradley,

Bill Winston, Karen E. DeMatteo, K. A. I. Nekaris, and Emily E. Wroblewski designed methodology. Leslie Annette Paige, Katherine Hedger, Esther Adinda, Endah Sulistyawati, and K. A. I. Nekaris collected the data. Leslie Annette Paige and Karen E. DeMatteo analysed the data. Leslie Annette Paige led the writing of the manuscript and Emily E. Wroblewski supervised the writing of the manuscript. All authors contributed critically to the drafts and gave final approval for publication.

ACKNOWLEDGEMENTS

Thank you to the many individuals that made this research possible. Special thanks go to the Little Fireface Projects's team of trackers: Rahmat Hidayat, Andri Yansah, Aldi Safari Nugraha, Pak Juhana, and Pak Aab Adurohman. Thanks to K. A. I. Nekaris, Katherine Hedger, and Endah Sulistyawati, as permit holders. Thank you to Garut District Perhutani, Garut District BKSDA, and the National Research and Innovation Agency (BRIN) of Indonesia for permission to conduct this research. Thank you to Dr. Michael Frachetti and the Spatial Analysis Interpretation and Exploration (SAIE) Lab at Washington University in St. Louis Department of Anthropology for computing support. Thank you to the Fossett Lab at Washington University in St. Louis Department of Earth, Environmental and Planetary Sciences for permission to use the drone. Thank you to Esther Adinda for translation of the abstract into Bahasa Indonesia. Thank you to this project's funders including Washington University in St. Louis, People's Trust for Endangered Species, Cleveland Metroparks Zoo and Augsburg Zoo.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Data including classified drone imagery, loris home 100% minimum convex polygon range outlines and summarized data derived from the drone and satellite imagery are available in Zenodo <https://doi.org/10.5281/zenodo.18246374> (Paige et al., 2026). Orthomosaic maps generated from the drone surveys will be available upon request due to privacy concerns given the resolution of the imagery. Loris ranging data will also be available upon request to remain compliant with Indonesian regulations. We do not have permission to share Planet imagery or associated primary outputs.

ORCID

Leslie Annette Paige  <https://orcid.org/0009-0002-5648-1344>

Karen E. DeMatteo  <https://orcid.org/0000-0002-3941-7538>

K. A. I. Nekaris  <https://orcid.org/0000-0001-5523-7353>

REFERENCES

Bertacchi, A., Giannini, V., Di Franco, C., & Silvestri, N. (2019). Using unmanned aerial vehicles for vegetation mapping and identification of botanical species in wetlands. *Landscape and Ecological Engineering*, 15(2), 231–240.

Biro, H., Campera, M., Imron, M. A., & Nekaris, K. A. I. (2020). Artificial canopy bridges improve connectivity in fragmented landscapes:

The case of Javan slow lorises in an agroforest environment. *American Journal of Primatology*, 82(4), e23076.

Blaschke, T. (2010). Object based image analysis for remote sensing. *ISPRS Journal of Photogrammetry and Remote Sensing: Official Publication of the International Society for Photogrammetry and Remote Sensing*, 65(1), 2–16.

Bonan, G. B. (2008). Forests and climate change: Forcings, feedbacks, and the climate benefits of forests. *Science*, 320(5882), 1444–1449.

Brandt, M., Tucker, C. J., Kariryaa, A., Rasmussen, K., Abel, C., Small, J., Chave, J., Rasmussen, L. V., Hiernaux, P., Diouf, A. A., Kergoat, L., Mertz, O., Igel, C., Gieseke, F., Schöning, J., Li, S., Melocik, K., Meyer, J., Sinno, S., ... Fensholt, R. (2020). Fensholt, unexpectedly large count trees West African Sahara Sahel. *Nature*, 587, 78–82.

Canadell, J. G., & Raupach, M. R. (2008). Managing forests for climate change mitigation. *Science (New York, N.Y.)*, 320(5882), 1456–1457.

Congalton, R. G. (2001). Accuracy assessment and validation of remotely sensed and other spatial information. *International Journal of Wildland Fire*, 10(4), 321.

Cruzan, M. B., Weinstein, B. G., Grasty, M. R., Kohn, B. F., Hendrickson, E. C., Arredondo, T. M., & Thompson, P. G. (2016). Small unmanned aerial vehicles (micro-UAVs, drones) in plant ecology. *Applications in Plant Sciences*, 4(9), 1600041. <https://doi.org/10.3732/apps.1600041>

Curtis, P. G., Slay, C. M., Harris, N. L., Tyukavina, A., & Hansen, M. C. (2018). Classifying drivers of global forest loss. *Science*, 361(6407), 1108–1111.

Durán, A. P., Green, J. M. H., West, C. D., Visconti, P., Burgess, N. D., Virah-Sawmy, M., & Balmford, A. (2020). A practical approach to measuring the biodiversity impacts of land conversion. *Methods in Ecology and Evolution*, 11(8), 910–921.

Estes, L., Chen, P., Debats, S., Evans, T., Ferreira, S., Kuemmerle, T., Ragazzo, G., Sheffield, J., Wolf, A., Wood, E., & Caylor, K. (2018). A large-area, spatially continuous assessment of land cover map error and its impact on downstream analyses. *Global Change Biology*, 24(1), 322–337.

Estrada, A., Garber, P. A., Rylands, A. B., Roos, C., Fernandez-Duque, E., Di Fiore, A., Nekaris, K. A. I., Nijman, V., Heymann, E. W., Lambert, J. E., Rovero, F., Barelli, C., Setchell, J. M., Gillespie, T. R., Mittermeier, R. A., Arregoitia, L. V., de Guinea, M., Gouveia, S., Dobrovolski, R., & Li, B. (2017). Impending extinction crisis of the world's primates: Why primates matter. *Science Advances*, 3(1), e1600946.

Foley, J. A., Defries, R., Asner, G. P., Barford, C., Bonan, G., Carpenter, S. R., Chapin, F. S., Coe, M. T., Daily, G. C., Gibbs, H. K., Helkowski, J. H., Holloway, T., Howard, E. A., Kucharik, C. J., Monfreda, C., Patz, J. A., Prentice, I. C., Ramankutty, N., & Snyder, P. K. (2005). Global consequences of land use. *Science (New York, N.Y.)*, 309(5734), 570–574.

Foody, G. M. (2002). Status of land cover classification accuracy assessment. *Remote Sensing of Environment*, 80(1), 185–201.

Gibril, M. B., Kalantar, B., Al-Ruzouq, R., Ueda, N., Saeidi, V., Shanableh, A., Mansor, S., & Shafri, H. (2020). Mapping heterogeneous urban landscapes from the fusion of digital surface model and unmanned aerial vehicle-based images using adaptive multiscale image segmentation and classification. *Remote Sensing*, 12, 1081.

Global Forest Watch. (2023). Indonesia. Global Forest Watch. <https://www.globalforestwatch.org/dashboards/country/IDN/>

Han, Y.-G., Yoo, S. H., & Kwon, O. (2017). Possibility of applying unmanned aerial vehicle (UAV) and mapping software for the monitoring of waterbirds and their habitats. *Journal of Ecology and Environment*, 41(1), 21. <https://doi.org/10.1186/s41610-017-0040-5>

Hossain, M. D., & Chen, D. (2019). Segmentation for object-based image analysis (OBIA): A review of algorithms and challenges from remote sensing perspective. *ISPRS Journal of Photogrammetry and Remote Sensing*, 150, 115–134.

- Hsu, A. J., Kumagai, J., Favoretto, F., Dorian, J., Guerrero Martinez, B., & Aburto-Oropeza, O. (2020). Driven by drones: Improving mangrove extent maps using high-resolution remote sensing. *Remote Sensing*, 12(23), 3986.
- Hurskainen, P., Adhikari, H., Siljander, M., Pellikka, P. K. E., & Hemp, A. (2019). Auxiliary datasets improve accuracy of object-based land use/land cover classification in heterogeneous savanna landscapes. *Remote Sensing of Environment*, 233, 111354.
- Illarionova, S., Tregubova, P., Shukhratov, I., Shadrin, D., Efimov, A., & Burnaev, E. (2024). Advancing forest carbon stocks' mapping using a hierarchical approach with machine learning and satellite imagery. *Scientific Reports*, 14(1), 21032.
- Jacobson, A. P., Riggio, J., M Tait, A., & E M Baillie, J. (2019). Global areas of low human impact ('low impact areas') and fragmentation of the natural world. *Scientific Reports*, 9(1), 14179.
- Jones, G. W. (2013). *The 2010–2035 Indonesian population projection: Understanding the causes, consequences and policy option for population and development*. UNFPA Indonesia.
- Kamarudin, K. R., & Wei, Y. J. (2022). Production of drone orthomosaic map of UTHM Wetland Conservation Research Station using UAV photogrammetry. *IOP Conference Series: Earth and Environmental Science*, 1064(1), 012012.
- Karimloo, L., Campera, M., Imron, M. A., Rakholia, S., Mehta, A., Hedger, K., & Nekaris, K. A. I. (2023). Habitat use, terrestriality and feeding behaviour of Javan slow lorises in urban areas of a multi-use landscape in Indonesia. *Land*, 12(7), 1349.
- Koh, L. P., & Wich, S. A. (2012). Dawn of drone ecology: Low-cost autonomous aerial vehicles for conservation. *Tropical Conservation Science*, 5(2), 121–132.
- Lambin, E. F., & Meyfroidt, P. (2011). Global land use change, economic globalization, and the looming land scarcity. *Proceedings of the National Academy of Sciences of the United States of America*, 108(9), 3465–3472.
- Lechner, A. M., Foody, G. M., & Boyd, D. S. (2020). Applications in remote sensing to forest ecology and management. *One Earth*, 2(5), 405–412.
- Lechner, A. M., Langford, W. T., Bekessy, S. A., & Jones, S. D. (2012). Are landscape ecologists addressing uncertainty in their remote sensing data? *Landscape Ecology*, 27(9), 1249–1261.
- Lechner, A. M., Stein, A., Jones, S. D., & Ferwerda, J. G. (2009). Remote sensing of small and linear features: Quantifying the effects of patch size and length, grid position and detectability on land cover mapping. *Remote Sensing of Environment*, 113(10), 2194–2204.
- Li, H., Di, L., Zhang, C., Lin, L., Guo, L., Yu, E. G., & Yang, Z. (2024). Automated in-season crop-type data layer mapping without ground truth for the conterminous United States based on multisource satellite imagery. *IEEE Transactions on Geoscience and Remote Sensing: A Publication of the IEEE Geoscience and Remote Sensing Society*, 62, 1–14.
- Lindenmayer, D. (2019). Small patches make critical contributions to biodiversity conservation. *Proceedings of the National Academy of Sciences of the United States of America*, 116(3), 717–719.
- Liu, S., Brandt, M., Nord-Larsen, T., Chave, J., Reiner, F., Lang, N., Tong, X., Ciais, P., Igel, C., Pascual, A., Guerra-Hernandez, J., Li, S., Mugabowindekwe, M., Saatchi, S., Yue, Y., Chen, Z., & Fensholt, R. (2023). The overlooked contribution of trees outside forests to tree cover and woody biomass across Europe. *Science Advances*, 9(37), eadh4097.
- Manohar, N., Pranav, M. A., Aksha, S., & Mytravarun, T. K. (2021). Classification of satellite images. In *Information and communication technology for intelligent systems* (pp. 703–713). IntechOpen.
- Martínez-Carricondo, P., Agüera-Vega, F., Carvajal-Ramírez, F., Mesas-Carrascosa, F.-J., García-Ferrer, A., & Pérez-Porras, F.-J. (2018). Assessment of UAV-photogrammetric mapping accuracy based on variation of ground control points. *International Journal of Applied Earth Observation and Geoinformation*, 72, 1–10.
- Masolele, R. N., Marcos, D., De Sy, V., Abu, I.-O., Verbesselt, J., Reiche, J., & Herold, M. (2024). Mapping the diversity of land uses following deforestation across Africa. *Scientific Reports*, 14(1), 1681.
- Mercier, A., Betbeder, J., Rumiano, F., Baudry, J., Gond, V., Blanc, L., Bourgoign, C., Cornu, G., Ciudad, C., Marchamalo, M., Pocard-Chapuis, R., & Hubert-Moy, L. (2019). Evaluation of Sentinel-1 and 2 time series for land cover classification of forest-agriculture mosaics in temperate and tropical landscapes. *Remote Sensing*, 11(8), 979.
- Mittermeier, R. A., Reuter, K. E., Rylands, A. B., Jerusalinsky, L., Schwitzer, C., Strier, K. B., Ratsimbazafy, J., & Humle, T. (2022). *Primates in peril: The world's 25 most endangered primates 2022–2023*. IUCN SSC Primate Specialist Group, International Primatological Society.
- Mokany, K., Ferrier, S., Harwood, T. D., Ware, C., Di Marco, M., Grantham, H. S., Venter, O., Hoskins, A. J., & Watson, J. E. M. (2020). Reconciling global priorities for conserving biodiversity habitat. *Proceedings of the National Academy of Sciences*, 117(18), 9906–9911.
- Mugabowindekwe, M., Brandt, M., Chave, J., Reiner, F., Skole, D. L., Kariryaa, A., Igel, C., Hiernaux, P., Ciais, P., Mertz, O., Tong, X., Li, S., Rwanyiziri, G., Dushimiyimana, T., Ndoli, A., Uwizeyimana, V., Lillesø, J.-P. B., Gieseke, F., Tucker, C. J., & Fensholt, R. (2023). Nation-wide mapping of tree-level aboveground carbon stocks in Rwanda. *Nature Climate Change*, 13(1), 91–97.
- Murugan, D., Garg, A., & Singh, D. (2017). Development of an adaptive approach for precision agriculture monitoring with drone and satellite data. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 10(12), 5322–5328.
- Nair, P. K. R., Kumar, B. M., & Nair, V. D. (2021). Global distribution of agroforestry systems. In *An Introduction to Agroforestry* (pp. 45–58). Springer International Publishing.
- Natural Resources Canada. (2017). *Canadian spatial reference system precise point positioning (CSRS-PPP)*. Government of Canada.
- Nekaris, K. A. I., Balestri, M., Hedger, K., Campera, M., Adinda, E., Ahmad, N., Imron, M. A., Manson, S., Saepurohman, F., & Nijman, V. (2024). Interactions between Javan slow lorises (*Nycticebus javanicus*) and domestic and wild carnivores in an anthropogenic landscape in Java, Indonesia. *Folia Primatologica; International Journal of Primatology*, 96, 1–15.
- Nekaris, K. A. I., Campera, M., Nijman, V., Birot, H., Rode-Margono, E. J., Fry, B. G., Weldon, A., Wirdateti, W., & Imron, M. A. (2020). Slow lorises use venom as a weapon in intraspecific competition. *Current Biology: CB*, 30(20), R1252–R1253.
- Nekaris, K. A. I., Poindexter, S., Reinhardt, K. D., Sigaud, M., Cabana, F., Wirdateti, W., & Nijman, V. (2017). Coexistence between Javan slow lorises (*Nycticebus javanicus*) and humans in a dynamic agroforestry landscape in West Java, Indonesia. *International Journal of Primatology*, 38(2), 303–320.
- Newbold, T., Hudson, L. N., Hill, S. L. L., Contu, S., Lysenko, I., Senior, R. A., Börger, L., Bennett, D. J., Choimes, A., Collen, B., Day, J., De Palma, A., Díaz, S., Echeverria-Londoño, S., Edgar, M. J., Feldman, A., Garon, M., Harrison, M. L. K., Alhusseini, T., & Purvis, A. (2015). Global effects of land use on local terrestrial biodiversity. *Nature*, 520(7545), 45–50.
- Onishi, M., & Ise, T. (2021). Explainable identification and mapping of trees using UAV RGB image and deep learning. *Scientific Reports*, 11(1), 1–15.
- Paige, L. A., Hedger, K., Adinda, E., Sulistyawati, E., Bradley, A., Winston, B., DeMatteo, K. E., Nekaris, K. A. I., & Wroblewski, E. E. (2026). Refining classification of complex agroforestry mosaic landscape from drone-based imagery: Implications for landscape management and conservation [data set]. *Zenodo*. <https://doi.org/10.5281/zenodo.18246374>
- Piel, A. K., Cruncheon, A., Knot, I. E., Chalmers, C., Fergus, P., Mulero-Pázmány, M., & Wich, S. A. (2021). Noninvasive technologies for

- primate conservation in the 21st century. *International Journal of Primatology*, 43, 1–25. <https://doi.org/10.1007/s10764-021-00245-z>
- Planet Team. (2017). *Planet application program interface: In space for life on earth*. Planet Team.
- Pravitasari, A. E., Indraprahasta, G. S., Rustiadi, E., Rosandi, V. B., Stanny, Y. A., Wulandari, S., Priatama, R. A., & Murtadho, A. (2024). Dynamics and predictions of urban expansion in Java, Indonesia: Continuity and change in mega-urbanization. *ISPRS International Journal of Geo-Information*, 13(3), 102.
- Pu, R., & Landry, S. (2012). A comparative analysis of high spatial resolution IKONOS and WorldView-2 imagery for mapping urban tree species. *Remote Sensing of Environment*, 124, 516–533.
- Purwaningsih, R., Sartohadi, J., & Setiawan, M. A. (2020). Trees and crops arrangement in the agroforestry system based on slope units to control landslide reactivation on volcanic foot slopes in Java, Indonesia. *Land*, 9(9), 327.
- Quattrini, G., Pesaresi, S., Hofmann, N., Mancini, A., & Casavecchia, S. (2025). Integrating drone truthing functional classification remote sensing time series supervised vegetation mapping. *Remote Sensing*, 17(2), 330.
- Radwan, T. M., Blackburn, G. A., Whyatt, J. D., & Atkinson, P. M. (2021). Global land cover trajectories and transitions. *Scientific Reports*, 11(1), 12814.
- Räsänen, A., & Virtanen, T. (2019). Data and resolution requirements in mapping vegetation in spatially heterogeneous landscapes. *Remote Sensing of Environment*, 230, 111207.
- Reiner, F., Brandt, M., Tong, X., Skole, D., Kariryaa, A., Ciais, P., Davies, A., Hiernaux, P., Chave, J., Mugabowindekwe, M., Igel, C., Oehmcke, S., Gieseke, F., Li, S., Liu, S., Saatchi, S., Boucher, P., Singh, J., Taourdeau, S., & Fensholt, R. (2023). More than one quarter of Africa's tree cover is found outside areas previously classified as forest. *Nature Communications*, 14(1), 2258.
- Santos, M., Cajaiba, R. L., Bastos, R., Gonzalez, D., Petrescu Bakis, A.-L., Ferreira, D., Leote, P., da Barreto Silva, W., Cabral, J. A., Gonçalves, B., & Mosquera-Losada, M. R. (2022). Why do agroforestry systems enhance biodiversity? Evidence from habitat amount hypothesis predictions. *Frontiers in Ecology and Evolution*, 9, 630151.
- Schaffer-Smith, D., Swenson, J. J., & Bóveda-Penalba, A. J. (2016). Rapid conservation assessment for endangered species using habitat connectivity models. *Environmental Conservation*, 43(3), 221–230.
- Schmitt, C. B., Burgess, N. D., Coad, L., Belokurov, A., Besançon, C., Boisrobert, L., Campbell, A., Fish, L., Gliddon, D., Humphries, K., Kapos, V., Loucks, C., Lysenko, I., Miles, L., Mills, C., Minnemeyer, S., Pistorius, T., Ravilious, C., Steininger, M., & Winkel, G. (2009). Global analysis of the protection status of the world's forests. *Biological Conservation*, 142(10), 2122–2130.
- Schnell, S., Kleinn, C., & Ståhl, G. (2015). Monitoring trees outside forests: A review. *Environmental Monitoring and Assessment*, 187(9), 600.
- Seifert, E., Seifert, S., Vogt, H., Drew, D., van Aardt, J., Kunneke, A., & Seifert, T. (2019). Influence of drone altitude, image overlap, and optical sensor resolution on multi-view reconstruction of forest images. *Remote Sensing*, 11(10), 1252.
- Semenchuk, P., Plutzar, C., Kastner, T., Matej, S., Bidoglio, G., Erb, K.-H., Essl, F., Haberl, H., Wessely, J., Krausmann, F., & Dullinger, S. (2022). Relative effects of land conversion and land-use intensity on terrestrial vertebrate diversity. *Nature Communications*, 13(1), 615.
- Stark, D. J., Vaughan, I. P., Evans, L. J., Kler, H., & Goossens, B. (2018). Combining drones and satellite tracking as an effective tool for informing policy change in riparian habitats: A proboscis monkey case study. *Remote Sensing in Ecology and Conservation*, 4(1), 44–52.
- Szantoi, Z., Smith, S. E., Strona, G., Koh, L. P., & Wich, S. A. (2017). Mapping orangutan habitat and agricultural areas using Landsat OLI imagery augmented with unmanned aircraft system aerial photography. *International Journal of Remote Sensing*, 38(8–10), 2231–2245.
- Thomas, N., Baltezar, P., Lagomasino, D., Stovall, A., Iqbal, Z., & Fatoyinbo, L. (2021). Trees outside forests are an underestimated resource in a country with low forest cover. *Scientific Reports*, 11(1), 7919.
- Turner, B. L., 2nd, Lambin, E. F., & Reenberg, A. (2007). The emergence of land change science for global environmental change and sustainability. *Proceedings of the National Academy of Sciences of the United States of America*, 104(52), 20666–20671.
- Udawatta, R. P., Rankoth, L. M., & Jose, S. (2021). Agroforestry for biodiversity conservation. In *Agroforestry and Ecosystem Services* (pp. 245–274). Springer International Publishing.
- Ventura, D., Napoleone, F., Cannucci, S., Alleaume, S., Valentini, E., Casoli, E., & Burrascano, S. (2022). Integrating low-altitude drone based imagery and OBIA for mapping and manage semi natural grassland habitats. *Journal of Environmental Management*, 321, 115723.
- Weinstein, B. G., Marconi, S., Bohlman, S. A., Zare, A., Singh, A., Graves, S. J., & White, E. P. (2021). A remote sensing derived data set of 100 million individual tree crowns for the national ecological observatory network. *eLife*, 10, e62922.
- Weiss, M., Jacob, F., & Duveiller, G. (2020). Remote sensing for agricultural applications: A meta-review. *Remote Sensing of Environment*, 236, 111402.
- Winkler, K., Fuchs, R., Rounsevell, M., & Herold, M. (2021). Global land use changes are four times greater than previously estimated. *Nature Communications*, 12(1), 2501.
- Wintle, B. A., Kujala, H., Whitehead, A., Cameron, A., Veloz, S., Kukkala, A., Moilanen, A., Gordon, A., Lentini, P. E., Cadenhead, N. C. R., & Bekessy, S. A. (2019). Global synthesis of conservation studies reveals the importance of small habitat patches for biodiversity. *Proceedings of the National Academy of Sciences of the United States of America*, 116(3), 909–914.
- Wu, B., Zhang, M., Zeng, H., Tian, F., Potgieter, A. B., Qin, X., Yan, N., Chang, S., Zhao, Y., Dong, Q., Boken, V., Plotnikov, D., Guo, H., Wu, F., Zhao, H., Deronde, B., Tits, L., & Loupian, E. (2023). Challenges and opportunities in remote sensing-based crop monitoring: A review. *National Science Review*, 10(4), nwac290.
- Xiang, H., & Tian, L. (2011). Development of a low-cost agricultural remote sensing system based on an autonomous unmanned aerial vehicle (UAV). *Biosystems Engineering*, 108(2), 174–190.
- Yan, Y., Jarvie, S., Zhang, Q., Zhang, S., Han, P., Liu, Q., & Liu, P. (2021). Small patches are hotspots for biodiversity conservation in fragmented landscapes. *Ecological Indicators*, 130, 108086.
- Young, D. J. N., Koontz, M. J., & Weeks, J. (2022). Optimizing aerial imagery collection and processing parameters for drone-based individual tree mapping in structurally complex conifer forests. *Methods in Ecology and Evolution*, 13(7), 1447–1463.
- Zhang, J., Hu, J., Lian, J., Fan, Z., Ouyang, X., & Ye, W. (2016). Seeing the forest from drones: Testing the potential of lightweight drones as a tool for long-term forest monitoring. *Biological Conservation*, 198, 60–69.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Table S1. Chi-squared outputs of LULC comparisons for the total surveyed area.

Table S2. Full confusion matrix for the home range of loris Group 1.

Table S3. Full confusion matrix for the home range of loris Group 2.

Table S4. Full confusion matrix for the home range of loris Group 3.

Table S5. Full confusion matrix for the home range of loris Group 4.

Table S6. Full confusion matrix for the home range of loris Group 4.

Table S7. Full confusion matrix for the home range of loris Group 6.

Table S8. Full confusion matrix for the home range of loris Group 7.

Table S9. Full confusion matrix for the home range of loris Group 8.

Table S10. Full confusion matrix for the home range of loris Group 9.

Table S11. Chi-squared outputs of Javan slow loris GPS waypoint classification comparison for the total surveyed area.

Figure S1. Percent area of habitat classifications for each loris group home range from the satellite (light blue) versus drone (dark blue) classifications.

Figure S2. Comparison of the proportion of classified GPS waypoints falling within each LULC type for each loris group from the satellite (light blue) versus drone (dark blue) classifications.

How to cite this article: Paige, L. A., Hedger, K., Adinda, E., Sulistyawati, E., Bradley, A. S., Winston, B., DeMatteo, K. E., Nekaris, K. A. I., & Wroblewski, E. E. (2026). Refining classification of complex agroforestry mosaic landscape from drone-based imagery: Implications for landscape management and conservation. *Journal of Applied Ecology*, 63, e70290. <https://doi.org/10.1111/1365-2664.70290>